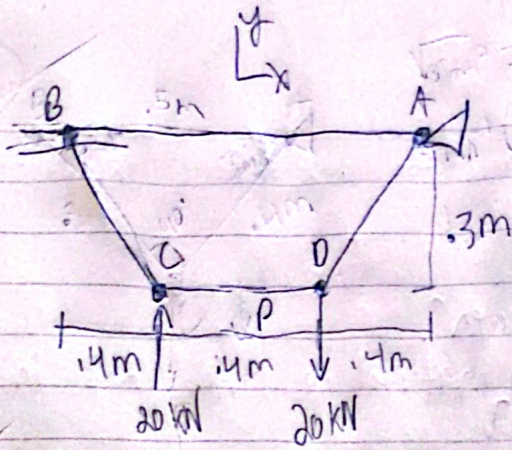
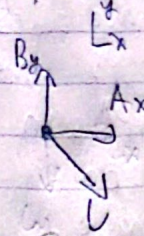


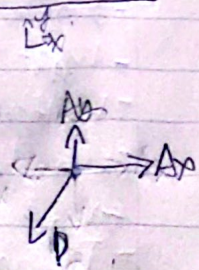
$$P = 20 \text{ kN}$$



FBD: B

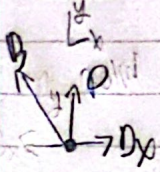


FBD: A

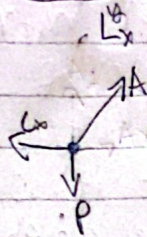


2

FBD C:



FBD D:



$$\sum M_A = -B_y(12) - 20(6.4) + 20(1.4) = 0$$

$$\sum F_y = B_y - 20 - 20 + 20 = 0$$

$$\sum F_x = -1.2 B_y = 0 \rightarrow B_y = -6.67 \text{ kN}$$

$$\sum F_y = A_y + B_y + 20 - 20 = 0$$

$$A_y - 6.67 = 0, A_y = 6.67 \text{ kN}$$

$$\sum F_y = 0$$

$$0.6 F_{BC} + 20 = 0$$

$$F_{BC} = -33.33 \text{ kN/C}$$

$$\sum F_y = 0, -0.8 F_{BC} + F_{CD} = 0$$

$$-0.8(-33.33) + F_{CD} = 0 \rightarrow -26.67 \text{ kN/C}$$

$$0.6 F_{AD} - 20 = F_{AD} = 33.33 \text{ kN} \uparrow$$

$$0.8(33.33) - F_{CD} = 0$$

$$F_{CD} = 26.67 \text{ kN}$$

3

$$L_{AB} = 12 \text{ m}, L_{BC} = 0.5 \text{ m}, L_{CD} = 1.4 \text{ m}, L_{DA} = 0.5 \text{ m}, L_{total} = 2.6 \text{ m}$$

unknown: A_{min} , W_{truss}

$$\sigma = \frac{F}{A}$$

$$FS = \frac{\sigma_y}{\sigma_{allow}}$$

$$\frac{F_{max}}{A_{min}} \leq \frac{\sigma_y}{FS}$$

$$A_{min} = \frac{FS F_{max}}{\sigma_y}$$

$$\sigma_{allow} = \frac{170}{3.5} = 48.57 \text{ ksi}$$

$$33.33 \text{ kN} \rightarrow 7.44 \text{ kip}$$

$$A_{min} = \frac{7.44}{48.57} = 0.154 \text{ in}^2 \rightarrow 99.5 \text{ mm}^2$$

$$A_{min} = \frac{VFS}{\tau_y}$$

$$V_{total} = (4 \text{ pins})(A_{min})(L_{pin})$$

$$\frac{(6670 \text{ N}) 4}{(17,000)} \left(\frac{1}{4448} \right) = A_{min} = 0.148 \text{ in}^2 = 90.9 \text{ mm}^2$$